A SYSTEM AND METHOD FOR DETECTING POTENTIALLY EVASIVE OR FRAUDULENT ENTITIES BY ANALYZING DATA ELEMENT TRANSPOSITION IN PUBLIC RECORDS

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Abstract

A data processing system and method for identifying entities that may be attempting to evade legal service, automated detection, or data correlation. The system operates by ingesting data records from one or more databases, such as public corporate registries, contact directories, or government databases. For a given data field containing a string of characters or numbers (e.g., a company name, a person's name, or a phone number), the system generates a set of permutations by systematically transposing adjacent or logically grouped elements within the string. These permuted variants are then used as search queries against the same or different databases. A successful query result for a permuted variant that differs

from the original record is flagged as a "transposition hit." The system aggregates these hits and calculates a risk score, identifying entities that are associated with multiple, deliberate variations of their data, a behavior indicative of an intent to obfuscate identity or evade detection.

Background of the Invention

In the digital age, entities are often identified and tracked across various databases by key identifiers such as names, addresses, and phone numbers. The accuracy and consistency of this data is critical for legal processes (e.g., service of process), regulatory compliance (e.g., know-your-customer), and data analysis.

Malicious or evasive actors often attempt to break this data consistency by introducing deliberate "errors" into their data as it is entered into different systems. This makes it difficult for automated systems to correlate their records and for human investigators to find all relevant information. A common, simple, yet effective technique is the transposition of elements within a data string. For example, an entity named "ABC Corp" might register itself as "ACB Corp" in another jurisdiction, or a phone number '(212) 555-1234' might be listed as '(212) 555-1243'. These changes are subtle enough to be missed by standard search algorithms but effective enough to thwart direct matching.

There is a need for an automated system that can proactively detect this specific pattern of obfuscation across large datasets to identify entities that are likely engaging in evasive or fraudulent behavior.

Summary of the Invention

The present invention provides a system and method for automatically detecting entities that use data element transposition as a form of obfuscation.

The system ingests a source data record associated with an entity. The record contains one or more data fields, such as a name or a contact number. The system's core logic, the Transposition Engine, selects a target field and generates a set of permuted variations of the data in that field. For a name, this may involve swapping the first and last names. For a phone number, this may involve swapping adjacent digits or digits within the same block.

These generated permutations are then submitted as queries to one or more target databases, which may include the source database or external public record databases. The system analyzes the results of these queries. A "transposition hit" is registered when a query for a permuted variant returns a positive match for a different, but related, entity or record.

The system scores and aggregates these hits. An entity that has a high number of transposition hits across multiple data fields is assigned a high risk score, indicating a high probability that the entity is deliberately using transposition to obfuscate its identity. This risk score can then be used to flag the entity for further investigation, enhanced due to diligence, or other risk mitigation measures.

Brief Description of the Drawings

FIG. 1 is a flowchart illustrating the overall data processing method, from ingesting a record to assigning a risk score.

FIG. 2 is a conceptual diagram illustrating the generation of permuted variants for different types of data fields (name and phone number).

Detailed Description of the Invention

Referring to the flowchart in **FIG. 1**, the method begins by ingesting a source data record for a target entity (101). The system then selects a specific data field from this record to analyze, for example, a contact phone number (102).

The core of the system is the Transposition Engine, which generates a set of logical permutations of the data from the selected field (103). As illustrated in **FIG. 2**, the permutation logic is adapted to the data type. For a name field (201), such as "John Smith," a permutation might involve transposing the first and last name elements to create "Smith John." For a numeric field (202), such as a phone number '(212) 555-1234', a permutation might involve transposing adjacent digits to create '(212) 555-1243'.

Returning to FIG. 1, each generated permutation is then used as a query against one or more target databases (104). The system analyzes the results to see if any of the permuted queries return a match (105).

If a query for a permuted variant returns a valid record that is distinct from the source record, it is flagged as a "transposition hit" (106). For example, if the original entity's fax number was '(212) 935-1000', and a query for '(212) 953-1000' returns a record for what appears to be the same or a related entity, a hit is registered.

The system aggregates all hits associated with the target entity (107) and applies a scoring algorithm to calculate a final risk score (108). The scoring algorithm may weigh hits on certain fields (e.g., legal department contact information) more heavily than others. A

high score indicates a pattern of deliberate data transposition, and the entity is flagged as potentially evasive or fraudulent.

Claims

What is claimed is:

1. A computer-implemented method for identifying a potentially evasive entity, comprising the steps of:

    - Ingesting a source data record associated with an entity, the record containing a data field with an original data string;
    - Generating, with a processor, a plurality of permuted data strings by systematically transposing elements within the original data string;
    - Submitting each of the permuted data strings as a query to at least one database;
    - Identifying a transposition hit when a query for a permuted data string returns a valid data record distinct from the source data record; and
    - Assigning a risk score to the entity based on the number and type of identified transposition hits.

2. The method of claim 1, wherein the original data string is a person's name, and the step of generating permuted strings includes transposing the first name and last name.

3. The method of claim 1, wherein the original data string is a telephone number, and the step of generating permuted strings includes transposing two adjacent digits.

4. The method of claim 1, wherein the risk score is increased if the data field corresponds to a legal contact information field.

5. A system for detecting data obfuscation, comprising:

    - A data ingestion module configured to retrieve an entity record from a database;
    - A transposition engine configured to:
        - select a data field from the entity record containing an original data string; and
        - generate a set of permuted data strings by reordering elements of the original data string;

- A query module configured to submit the set of permuted data strings as search queries to a database; and
- a risk analysis module configured to:
  - identify a positive search result for a permuted data string as a transposition hit; and
  - calculate a risk score for the entity based on an aggregation of identified transposition hits.

6. The system of claim 5, wherein the transposition engine is configured to apply different permutation rules based on the data type of the selected data field.

Drawings

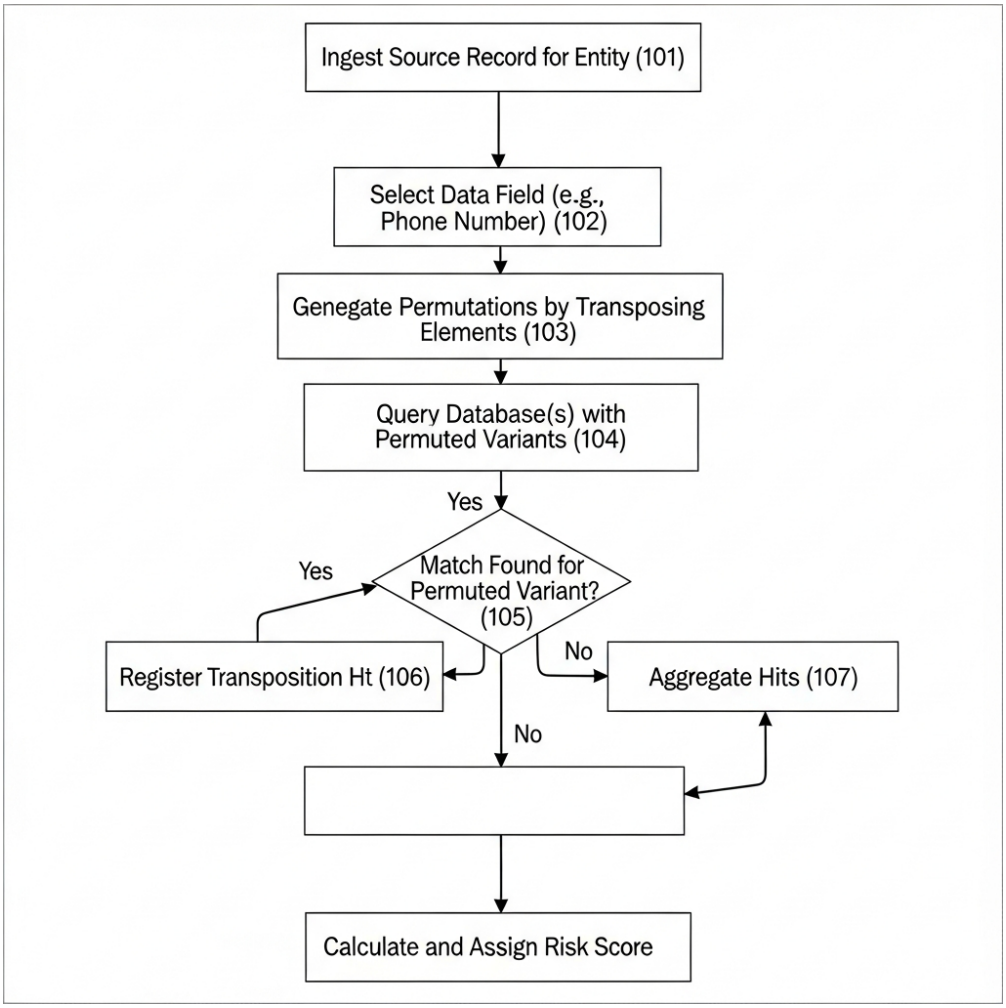


Figure 1: Flowchart of the data transposition detection method.

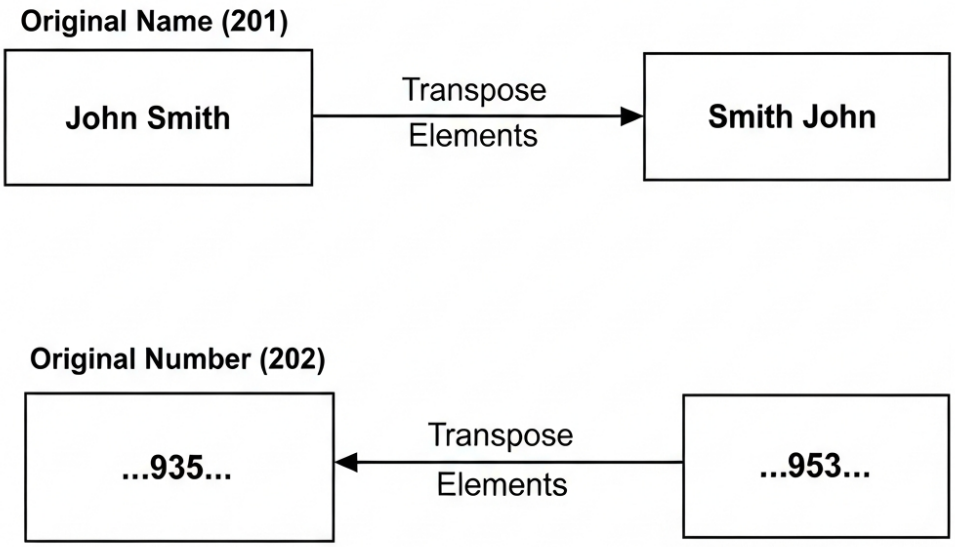


Figure 2: Conceptual diagram of permutation generation.